

Antony Bush

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Professional Summary

Backend Software Engineer specializing in scalable distributed systems, secure fintech infrastructure, and real-time communication. Proven track record of simplifying complex architectures, implementing zero-trust security, and driving high-impact optimizations, including reducing cloud costs by 23% and boosting service availability by 75%.

Professional Experience

Software Engineer, BondsIndia

Dec 2025 – Present

- **Architected** a maintainable backend for a product revamp, simplifying site architecture and enabling rapid deployment of new features without altering the core database schema.
- **Streamlined** the KYC onboarding module by integrating DigiLocker APIs, eliminating **two manual verification steps** and accelerating user approvals by **over 40%**.
- **Spearheaded** core database design focused on a zero-trust architecture, ensuring maximum data security and integrity across **100% of business services**.
- **Implemented** robust authentication and authorization flows handling **all API requests**, utilizing Keycloak for identity management and Kong Gateway for secure routing.
- **Developed** the IPO and Security Master modules, rapidly translating complex financial requirements into **scalable independent services**.

Software Engineer II, R3 Corda

Sept 2025 – Nov 2025

- **Refactored** Service Account Gateway toward Clean Architecture, decoupling Ktor from the library to make it platform-agnostic and improve testability/maintainability by 30%.
- **Orchestrated** Open Policy Agent authorization, introducing a zero-trust, security-first access model and delivering the first reference integration for Central Bank Digital Currency (CBDC) services, establishing the security standard for platform-wide implementation.

Software Development Engineer I, Jio Platforms Limited

Oct 2023 – Sept 2025

- **Engineered** the migration of business-critical microservices from Python to Node.js, standardizing implementation and achieving a **75% increase in service availability** and an **80% boost in request-handling capacity**.
- **Optimized** media server scalability for JioMeet & JioEvents, enabling **2× participants per node** and **23% cloud cost savings** by tuning SFU scaling logic.
- **Designed** a scalable chat microservice for JioMeet Enterprise, enabling advanced features (threaded replies, sharing) that lifted **user engagement metrics across 1M+ users**.
- **Automated** the complete Media Engine Stack server setup and CI/CD pipeline using Ansible, **slashing deployment time by 60%** and improving release reliability for engineering workflows.
- **Overhauled** recording and streaming services with performance optimizations, achieving a **30% reduction in CPU/memory usage** while enhancing stream quality for enterprise clients.
- **Pioneered** MCU capabilities alongside existing SFU architecture, improving stream delivery efficiency by 40% and **reducing client-side bandwidth by up to 60%** for 500+ participant meetings.
- **Integrated** LiveKit media infrastructure for JioMeet to **support real-time AI media features**, establishing the foundation for advanced AI-driven media processing at scale.

Skills

- **Languages & Frameworks:** TypeScript, JavaScript (Node.js, NestJS, Fastify, React)
- **Backend & Systems:** Ktor, Keycloak, Kong API Gateway, WebRTC, LiveKit, Open Policy Agent (OPA), RabbitMQ
- **DevOps & Cloud:** Docker, Kubernetes, Ansible, Terraform, Jenkins, CI/CD, Nginx, GCP, Linux, Bash
- **Databases & Caching:** PostgreSQL, MongoDB, Redis, MySQL
- **Monitoring & Observability:** Prometheus, Grafana, Elasticsearch

Education

B.E. in Computer Engineering, Mumbai University

July 2019 – July 2023

- CGPA: 9.67/10.0